

BIOINFORMATICS HACKATHON · 2026

Everything OMOP

Federated learning on harmonised biobank data

SUBPROJECT 1

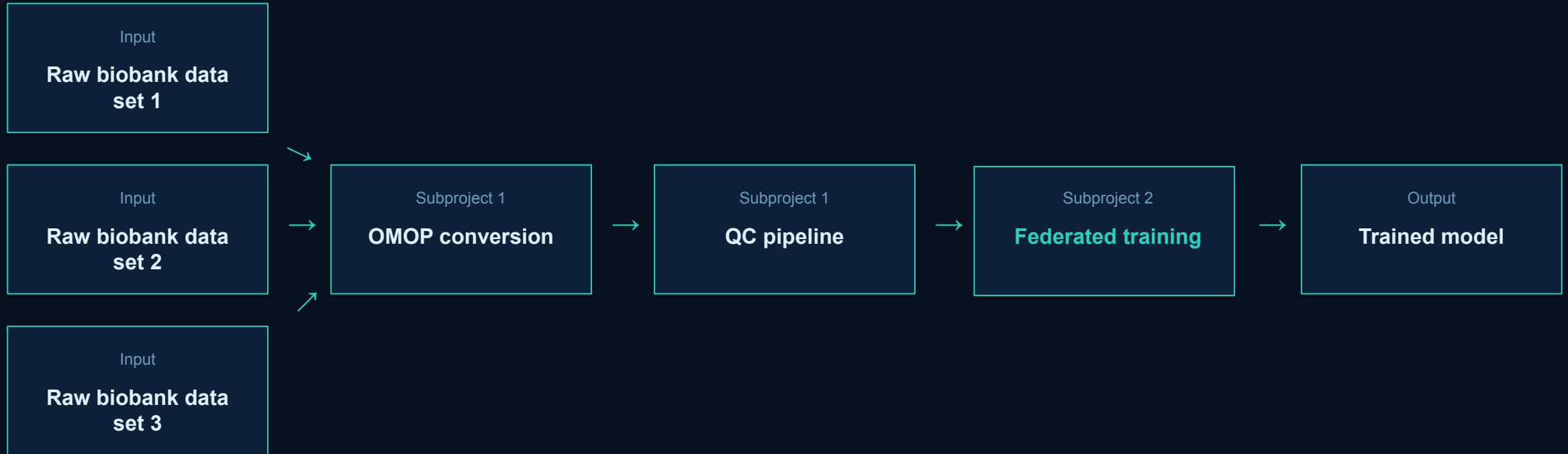
Automated ETL :
Raw Data/Biobank → OMOP conversion and quality control

SUBPROJECT 2

Proof-of-concept federated learning on OMOP datasets

OVERVIEW

End-to-end pipeline



OMOP conversion & quality control

in principle

Every biobank stores its data differently

OMOP makes analysis portable across sites

Status Quo: hand-built ETL per source, months of work

Our approach: an agent skill that converts, not a script per dataset

Scope: minimal table set, synthetic data, measurable

SUBPROJECT 1

OMOP conversion & quality control

CONVERSION PIPELINE

- Profile the source: fields, codes, distributions
- Propose structural mapping: source field → OMOP field
- Propose vocabulary mapping: source code → standard concept
- Clinical sign-off before a mapping is accepted
- Run ETL: contracted tables, one directory per site
- Sprechtext: Die ersten drei Schritte macht der Skill. Schritt vier ist bewusst ein Mensch. Schritt fünf ist reiner Code.
-

QUALITY CONTROL

Reference: published expert mappings (EHDEN, The Hyve)

- Metric 1: expert mappings recovered, correctly and wrongly
- Metric 2: share of records on concept_id = 0
- Metric 3: ten most frequent unmapped codes
- Output handed to subproject 2 via a fixed data contract

SUBPROJECT 2

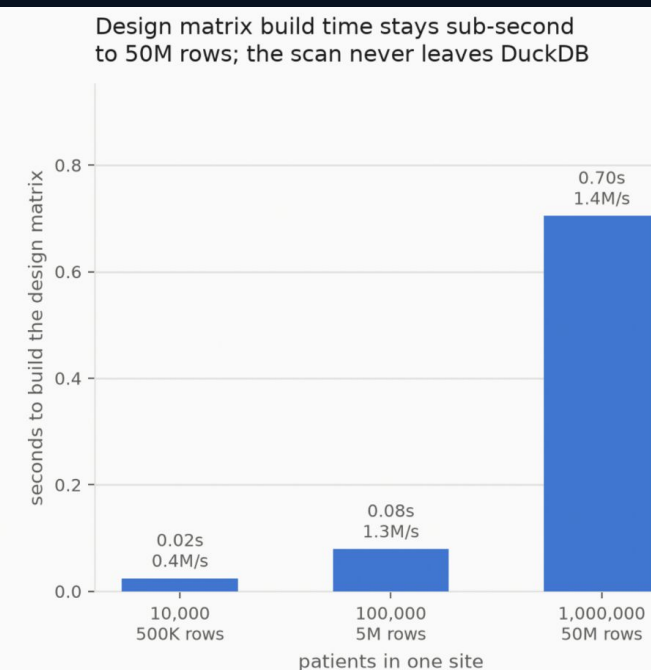
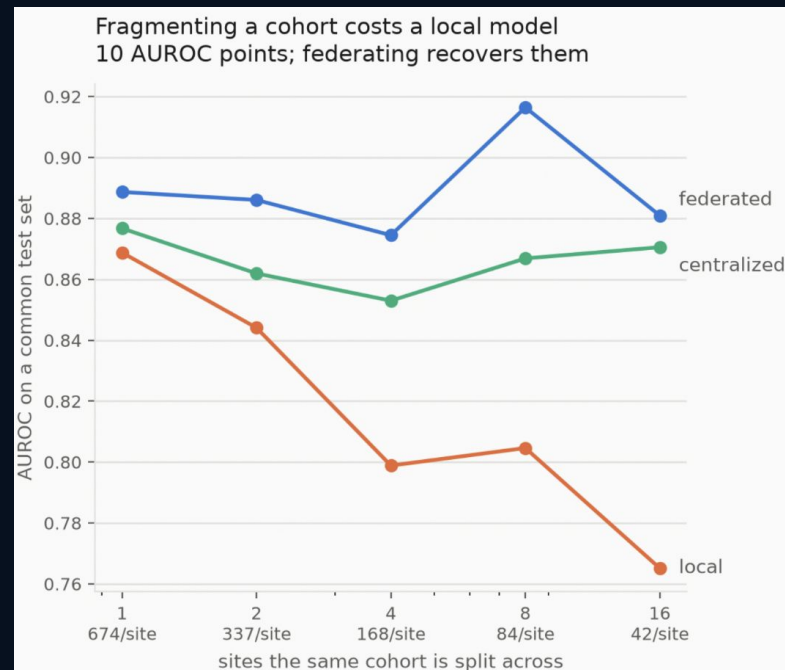
Federated learning on OMOP data

IMPLEMENTATION

- DuckDB enables lazy loading. Time-series data is stored as 3D pydata sparse arrays.
- Custom PyTorch dataloader & infrastructure is validated on 5 Synthea OMOP cohorts (3 newly generated & Eunomia GIBleed & Synthea27Nj). Validation on Hunt & synthetic UKB planned.
- Package “omopflare” provides an API to use all of our components. Integrates with nvflare.

SCOPE & OBJECTIVE

- Cox proportional hazard model trained across simulated federated sites
- Goal is [end-to-end pipeline functionality](#), not predictive performance
- Directly consumes OMOP output from Subproject 1



Federated learning on OMOP data

Proof of concept on simulated data

Biomarkers
BMI
SBP
HgA1c
Glucose
Creatinine
eGFR

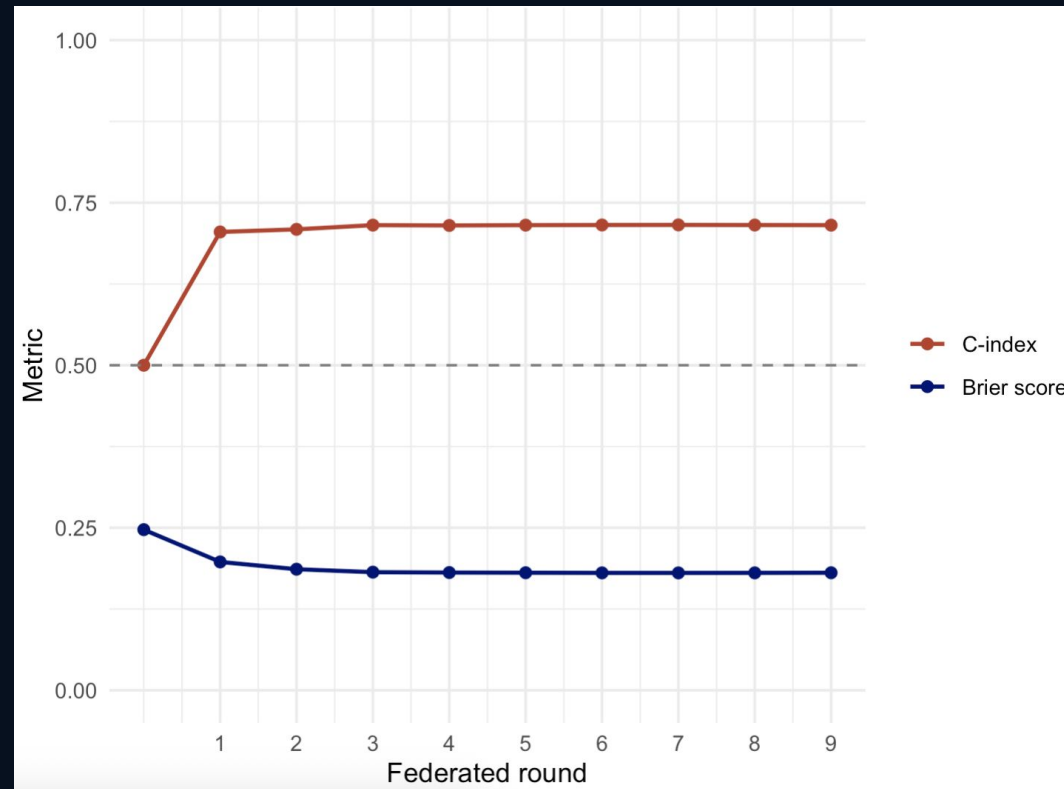
- Cox proportional hazards model predicting 5-year survival based on
 - Age
 - Blood based biomarkers
- Federation framework in place
 - Single cohort for the first demo

SUBPROJECT 2

Federated learning on OMOP data

Proof of concept on simulated data

Biomarkers
BMI
SBP
HgA1c
Glucose
Creatinine
eGFR



SUBPROJECT 2

Federated learning on OMOP data

Proof of concept on simulated data

Biomarkers
BMI
SBP
HgA1c
Glucose
Creatinine
eGFR

